SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Data Interpretation

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO4 – Precision	**Assessment:**	Assessment Tool 1 – Data Interpretation
Weightage:	35% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	Slot D / 3rd year / 2024-28	**Date:**	26/08/2026

Code of Conduct

I Shaik Mohammed Abrar / 192472185 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *Shaik Mohammed Abrar*

Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate	Q1

	meaning representation in chatbot systems.	
First-Order Predicate Calculus & Representing Linguistically Relevant Concepts	Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.	Q3
Syntax-Driven Semantic Analysis	Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.	Q4

Annexure A – Data Interpretation

1. Frame Semantics in Banking Customer-Service Systems

A digital banking platform uses semantic frames to understand customer requests. The NLP system identifies the **action, agent, object, and destination** involved in each request.

Customer Query	Generated Semantic Frame
"Transfer ₹20,000 from my savings account to my son's account."	TRANSFER(SourceAccount, Amount, DestinationAccount, Customer)
"Block my debit card immediately."	BLOCK(DebitCard, Customer)
"Send my transaction statement to my email."	SEND(TransactionStatement, Email, Customer)
"Increase my daily withdrawal limit."	MODIFY(WithdrawalLimit, Customer)

Additional transaction logs show:

Query ID	Actual User Intent	System Interpretation
B1	Transfer money to another account	Transfer money
B2	Block debit card	Block debit card
B3	Receive transaction statement by email	Send statement to email
B4	Increase withdrawal limit	Decrease withdrawal limit

Tasks:

1. Analyze the semantic frame of each banking request and identify the relationship between the action and its participants.
2. Identify the query in which the semantic interpretation is incorrect and explain why.
3. Analyze how incorrect identification of semantic roles can lead to an incorrect banking operation.
4. Recommend a frame-based semantic analysis approach to improve the reliability of banking conversational systems.

2. First-Order Predicate Calculus for Smart Agriculture

A smart agricultural system monitors crop fields using sensors and logical rules.

Field Data:

Field	Soil Condition	Irrigation Status	Crop
F1	Dry	Available	Rice
F2	Wet	Available	Wheat
F3	Dry	Unavailable	Maize
F4	Wet	Unavailable	Rice

The system uses the following predicate rules:

- $Dry(x) \land IrrigationAvailable(x) \rightarrow NeedsWater(x)$
- $NeedsWater(x) \rightarrow Irrigate(x)$

- Wet(x) → ¬NeedsWater(x)
- IrrigationUnavailable(x) → ¬Irrigate(x)
- Rice(x) ∧ NeedsWater(x) → PriorityIrrigation(x)

Tasks:

1. Represent the field observations using First-Order Predicate Calculus.
2. Using the given rules, determine which fields require irrigation.
3. Determine whether F1 and F3 should be treated differently by the automated irrigation controller.
4. Analyze whether predicate logic alone is sufficient for agricultural decision support when sensor information is incomplete or uncertain.

3. Word Sense Disambiguation in a Travel Recommendation System

A travel platform receives search queries containing ambiguous words. The recommendation engine uses surrounding words and user interactions to determine the intended meaning.

Search Query	Possible Sense 1	Possible Sense 2
"Amazon tour"	Amazon rainforest	Amazon company
"Safari booking"	Wildlife tour	Web browser
"Java vacation"	Java island	Java programming language
"Cruise package"	Ship journey	Cruise software/project

The system records the following user interactions:

Query	User Interaction
Amazon tour	Viewed rainforest trekking packages
Safari booking	Selected wildlife resort packages
Java vacation	Viewed hotels in Bali and Java
Cruise package	Viewed Mediterranean ship packages

However, another user searches: **"Safari download for Windows"** and clicks a browser download page.

Tasks:

1. Determine the intended sense of the ambiguous terms using the available context.
2. Identify the lexical and contextual cues that distinguish the possible senses.
3. Analyze why the same word may require different interpretations for different users.
4. Design an industrial-scale WSD strategy using query context, user history, embeddings, and click behaviour to improve travel recommendations.

4. Syntax-Driven Semantic Analysis in Legal Document Processing

A legal NLP system analyzes sentences from contracts and assigns semantic roles based on their syntactic structures.

The system produces the following analyses:

Legal Sentence	Parse Structure	Generated Semantic Interpretation
"The supplier delivered the	Subject–Verb–	Supplier = Agent, Equipment =

Legal Sentence	Parse Structure	Generated Semantic Interpretation
equipment to the buyer."	Object	Object, Buyer = Recipient
"The buyer received the equipment from the supplier."	Subject–Verb–Object	Buyer = Agent, Equipment = Object, Supplier = Recipient
"The company terminated the contract after the violation."	Subject–Verb–Object	Company = Agent, Contract = Object
"The contract was terminated by the company."	Passive Construction	Contract = Agent, Company = Object

Tasks:

1. Analyze how syntactic structure, including active and passive constructions, influences semantic-role assignment.
2. Identify the incorrect semantic-role assignments in the given interpretations.
3. Explain how confusing grammatical subject with semantic agent can affect automated legal information extraction.
4. Recommend a syntax-driven semantic analysis architecture that can correctly handle active voice, passive voice, prepositional phrases, and long-distance dependencies in legal documents.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions

Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria Level	Understanding of Data	Analysis	Application of Concepts	Conclusion	Clarity	Sub. Total	Total %
1	3	3	3	3	3	15	78
2	3	4	3	3	3	16	
3	3	3	4	3	3	16	
4	3	3	3	3	3	15	

Faculty In-charge	Course Coordinator	Program Director
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Date: 26/08/2026	Date: ____________________	Date: ____________________

Question 1 – Frame Semantics in Banking Customer-Service Systems

The banking system uses semantic frames to identify the action, participants and relevant entities in customer requests. The supplied frames are TRANSFER, BLOCK, SEND and MODIFY.

Semantic Frame Analysis

Customer Request	**Semantic Frame**	**Action**	**Participants / Roles**
Transfer 20,000 from my savings account to my son's account.	TRANSFER(SourceAccount, Amount, DestinationAccount, Customer)	Transfer	Source = savings account; Amount = ₹20,000; Destination = son's account; Customer = requester
Block my debit card immediately.	BLOCK(DebitCard, Customer)	Block	Object = debit card; Customer = requester
Send my transaction statement to my email.	SEND(TransactionStatement, Email, Customer)	Send	Object = transaction statement; Destination = email; Customer = requester
Increase my daily withdrawal limit.	MODIFY(WithdrawalLimit, Customer)	Increase / modify	Object = withdrawal limit; Customer = requester

Interpretation Error

The incorrect interpretation is Query B4. The actual user intent is 'Increase withdrawal limit', whereas the system interpretation is 'Decrease withdrawal limit'.

Query ID	**Actual Intent**	**System Interpretation**	**Status**
B1	Transfer money to another account	Transfer money	Correct
B2	Block debit card	Block debit card	Correct
B3	Receive transaction statement by email	Send statement to email	Correct
B4	Increase withdrawal limit	Decrease withdrawal limit	Incorrect

Why B4 Is Incorrect

1. The action direction is reversed: 'increase' means raise the allowed withdrawal amount, while 'decrease' means lower it.
2. The object, withdrawal limit, is correctly identified, but the intended modification operation is wrong.
3. The error is semantic rather than merely syntactic because the system assigns the opposite meaning to the requested operation.

Risk of Incorrect Semantic Roles

In a banking conversational system, an incorrect semantic interpretation can directly result in an incorrect banking operation. For B4, interpreting 'increase' as 'decrease' could reduce the customer's withdrawal limit when the customer explicitly requested a higher limit. Similar role errors in a transfer frame could potentially confuse the source account and destination account.

Recommended Frame-Based Approach

1. Detect the intent/action using a frame classifier.
2. Extract frame participants such as source account, destination account, amount and customer.
3. Represent direction-sensitive actions explicitly, for example INCREASE rather than a generic MODIFY label.
4. Validate extracted roles against the expected frame schema.
5. Use confirmation for high-impact banking operations when confidence is low.
6. Keep the semantic frame separate from the final transaction execution layer.

```
Customer Query
    |
    v
Intent / Frame Detection
    |
    v
Role & Entity Extraction
    |
    v
Frame Validation
    |
    v
```

```
Confidence Check
     |
     +---- Low confidence ---> User Confirmation
     |
     v
Banking Operation
```

Question 2 – First-Order Predicate Calculus for Smart Agriculture

The agricultural system monitors soil condition, irrigation availability and crop type. The supplied rules determine whether a field needs water and whether irrigation should be performed.

Predicate Representation

Field	Predicate Representation
F1	Dry(F1) ∧ IrrigationAvailable(F1) ∧ Rice(F1)
F2	Wet(F2) ∧ IrrigationAvailable(F2) ∧ Wheat(F2)
F3	Dry(F3) ∧ IrrigationUnavailable(F3) ∧ Maize(F3)
F4	Wet(F4) ∧ IrrigationUnavailable(F4) ∧ Rice(F4)

Given Logical Rules

```
1. Dry(x) ∧ IrrigationAvailable(x) → NeedsWater(x)
2. NeedsWater(x) → Irrigate(x)
3. Wet(x) → ¬NeedsWater(x)
4. IrrigationUnavailable(x) → ¬Irrigate(x)
5. Rice(x) ∧ NeedsWater(x) → PriorityIrrigation(x)
```

Field-by-Field Inference

F1 – Dry, Irrigation Available, Rice

```
Dry(F1)
IrrigationAvailable(F1)
--------------------------------
NeedsWater(F1)

NeedsWater(F1)
--------------------------------
```

Irrigate(F1)

Rice(F1) ∧ NeedsWater(F1)

PriorityIrrigation(F1)

Therefore, F1 requires irrigation and receives priority irrigation because it is a rice field.

F2 – Wet, Irrigation Available, Wheat

Wet(F2)

¬NeedsWater(F2)

F2 does not require irrigation because its soil is wet.

F3 – Dry, Irrigation Unavailable, Maize

Dry(F3) ∧ IrrigationUnavailable(F3)

Dry(F3) alone is not sufficient for rule 1 because
IrrigationAvailable(F3) is false.

IrrigationUnavailable(F3)

¬Irrigate(F3)

F3 is dry, but the supplied rules do not derive NeedsWater(F3) because irrigation availability is a required condition of rule 1. The controller also derives ¬Irrigate(F3) from irrigation unavailability.

F4 – Wet, Irrigation Unavailable, Rice

Wet(F4)

¬NeedsWater(F4)

IrrigationUnavailable(F4)

¬Irrigate(F4)

F4 does not require irrigation because it is wet, and irrigation is also unavailable.

Final Decision Table

Field	Needs Water?	Irrigation?	Priority Irrigation?
F1	Yes	Yes	Yes – Rice
F2	No	No	No
F3	Not derived from rule 1	No – unavailable	No
F4	No	No – unavailable	No

F1 vs F3

F1 and F3 both have dry soil, but their irrigation statuses differ. F1 has irrigation available, so the conjunction Dry(F1) ∧ IrrigationAvailable(F1) satisfies the first rule and leads to NeedsWater(F1), Irrigate(F1), and PriorityIrrigation(F1). F3 has irrigation unavailable, so the first rule cannot fire and the fourth rule explicitly gives ¬Irrigate(F3). Therefore, the controller must treat F1 and F3 differently.

Is Predicate Logic Alone Sufficient?

Predicate logic is useful for explicit rules and deterministic inference, but it is not sufficient by itself when sensor information is incomplete or uncertain. The supplied predicates treat observations as true or false. Real sensors can produce noisy, missing, delayed or uncertain measurements.

1. Missing sensor readings cannot always be represented reliably as simple true/false predicates.
2. Soil moisture may be measured with uncertainty rather than as simply Dry or Wet.
3. Probabilistic or fuzzy reasoning can represent confidence and gradual conditions.
4. A practical decision-support system can combine logical rules with probabilistic sensor models.

Question 3 – Word Sense Disambiguation in a Travel Recommendation System

The travel platform receives ambiguous queries and uses surrounding context and user interaction to determine the intended sense. The supplied examples include Amazon, Safari, Java and Cruise.

Sense Identification

Query	Possible Senses	Available Evidence	Intended Sense
Amazon tour	Amazon rainforest / Amazon company	Viewed rainforest trekking packages	Amazon rainforest
Safari booking	Wildlife tour / Web browser	Selected wildlife resort packages	Wildlife tour
Java vacation	Java island / Java programming language	Viewed hotels in Bali and Java	Java island

Cruise package	Ship journey / Cruise software/project	Viewed Mediterranean ship packages	Ship journey
Safari download for Windows	Wildlife tour / Web browser	Clicked a browser download page	Web browser

Contextual and Lexical Cues

Ambiguous Term	Travel Cue	Technology Cue
Amazon	tour, rainforest, trekking	company, online shopping
Safari	booking, wildlife, resort	download, Windows, browser
Java	vacation, hotels, Bali	programming, code, software
Cruise	package, Mediterranean, ship	software/project terminology

Why the Same Word Has Different Meanings

A lexical item can have multiple senses, and the intended sense depends on the query and user context. For example, 'Safari' in 'Safari booking' is associated with wildlife travel, while 'Safari download for Windows' contains strong software/browser cues. Therefore, a WSD system should not assign one permanent meaning to every occurrence of a word.

Industrial-Scale WSD Strategy

```
Query Text
  |
  +---- Context Features
  |       - neighboring words
  |       - query entities
  |
  +---- User Features
  |       - previous searches
  |       - clicks
  |       - viewed categories
  |
  +---- Semantic Features
  |       - contextual embeddings
  |       - candidate sense representations
  |
  v
```

```
Sense Candidate Generation
   |
   v
Contextual Ranking Model
   |
   v
Predicted Sense
   |
   v
Travel Recommendation / Search Ranking
   |
   v
Feedback from clicks and interactions
```

1. Use the complete query rather than an isolated ambiguous word.
2. Encode surrounding words with contextual embeddings.
3. Use user history and previously viewed categories when permitted.
4. Include click behavior as implicit relevance feedback.
5. Maintain candidate senses for ambiguous terms.
6. Train and evaluate the ranking model on query–sense or click-labeled data.
7. Continuously monitor errors and update the model with new query patterns.

Question 4 – Syntax-Driven Semantic Analysis in Legal Document Processing

The legal NLP system assigns semantic roles using syntactic structures. The supplied examples include active voice and passive voice. The key distinction is that grammatical subject and semantic agent are not always the same.

Sentence 1 – Active Voice

Sentence: 'The supplier delivered the equipment to the buyer.'

Element	**Role**
The supplier	Agent
the equipment	Object / Theme
the buyer	Recipient

The supplied interpretation is correct because the supplier performs the delivery action, the equipment is the transferred object, and the buyer receives it.

Sentence 2 – Active Voice with 'from'

Sentence: 'The buyer received the equipment from the supplier.'

The supplied interpretation incorrectly assigns Buyer = Agent and Supplier = Recipient.

Element	**Correct Role**	**Reason**
The buyer	Recipient	The buyer receives the equipment
the equipment	Object / Theme	The item being received
the supplier	Source	The supplier is introduced by 'from' and is the source of the equipment

Although the buyer is the grammatical subject, the buyer is not the Agent of the transfer event. The supplier is the source, while the buyer is the recipient.

Sentence 3 – Active Voice

Sentence: 'The company terminated the contract after the violation.'

Element	**Role**
The company	Agent
the contract	Object / Theme
after the violation	Temporal / causal context as expressed by the sentence

The supplied Company = Agent and Contract = Object interpretation is correct.

Sentence 4 – Passive Voice

Sentence: 'The contract was terminated by the company.'

The supplied interpretation is incorrect because it assigns Contract = Agent and Company = Object.

Element	**Correct Role**	**Explanation**
The contract	Object / Theme	It undergoes the termination action
the company	Agent	The 'by' phrase identifies the

		entity performing the action

In passive constructions, the grammatical subject can correspond to the semantic object or patient. The agent can appear in a 'by' prepositional phrase.

Incorrect Interpretations

Sentence	Given Interpretation	Correct Interpretation
The buyer received the equipment from the supplier.	Buyer = Agent; Supplier = Recipient	Buyer = Recipient; Supplier = Source
The contract was terminated by the company.	Contract = Agent; Company = Object	Contract = Object/Theme; Company = Agent

Why Subject ≠ Agent

A grammatical subject is a syntactic role, whereas an Agent is a semantic role. In active voice, the subject frequently performs the action, so the two can align. In passive voice, the entity affected by the action can become the grammatical subject, while the Agent appears in a 'by' phrase. Confusing these roles can lead a legal information-extraction system to reverse who performed an action and who was affected by it.

Recommended Syntax-Driven Architecture

```
Legal Document
   |
   v
Sentence Segmentation
   |
   v
Tokenization + POS Tagging
   |
   v
Dependency / Constituency Parsing
   |
   v
Voice Detection
(active / passive)
   |
   v
```

```
Semantic Role Labeling
Agent / Patient / Theme / Recipient / Source
      |
      v
Coreference + Entity Linking
      |
      v
Legal Event Representation
      |
      v
Information Extraction / Search
```

1. Use dependency parsing to identify subject, object and prepositional attachments.
2. Detect passive constructions so that the grammatical subject is not automatically labeled Agent.
3. Use semantic-role labeling to map syntactic relations to Agent, Patient/Theme, Recipient and Source.
4. Handle prepositional phrases such as 'to', 'from' and 'by' explicitly.
5. Use coreference resolution for references such as 'it', 'the party' and 'the supplier'.
6. Use document-level context for long-distance dependencies and cross-sentence relationships.
7. Represent extracted legal events in a structured form for downstream retrieval and reasoning.

Conclusion

The four questions demonstrate how meaning can be represented and interpreted at different levels of NLP. Frame semantics organizes banking requests around actions and participants, predicate calculus supports logical reasoning over agricultural field conditions, WSD resolves ambiguous lexical senses using context and interaction evidence, and syntax-driven semantic analysis maps legal sentence structure to semantic roles.

The analysis also highlights the consequences of incorrect interpretation. A reversed banking action can lead to the wrong operation, an incorrect predicate inference can cause inappropriate irrigation decisions, an incorrect word sense can produce irrelevant travel recommendations, and incorrect semantic roles can reverse the participants of a legal event. Therefore, reliable NLP systems must combine accurate representation, contextual interpretation, validation and appropriate reasoning.